

Homework 6 Solutions

BSTA 551

Problem 1: Neyman-Pearson Lemma (Lesson 15)

Adverse events in a clinical trial are monitored over successive reporting windows. The number of adverse events per window follows $X_i \stackrel{iid}{\sim} \text{Poisson}(\lambda)$. A safety board tests:

$$H_0 : \lambda = 2 \quad \text{vs} \quad H_a : \lambda = 4 \quad (n = 6 \text{ windows})$$

- (a) Compute $\Lambda^* = L(\lambda = 4)/L(\lambda = 2)$ and show the NP rejection region simplifies to $\sum X_i \geq c$.
 - (b) Find critical value c with $\alpha^* \leq 0.05$ using `ppois()`.
 - (c) Compute the power at $\lambda = 4$.
 - (d) Explain why the same rejection region is UMP for $H_0 : \lambda \leq 2$ vs $H_a : \lambda > 2$.
-

Part (a) — NP likelihood ratio

The joint PMF is $L(\lambda) = e^{-6\lambda} \lambda^{\sum x_i} / \prod x_i!$. The NP ratio is:

$$\Lambda^* = \frac{L(4)}{L(2)} = \frac{e^{-24} \cdot 4^{\sum x_i}}{e^{-12} \cdot 2^{\sum x_i}} = e^{-12} \cdot 2^{\sum x_i}$$

The rejection region $\Lambda^* \geq k$ becomes:

$$e^{-12} \cdot 2^{\sum x_i} \geq k \implies \sum x_i \cdot \ln 2 \geq \ln k + 12 \implies \sum_{i=1}^6 X_i \geq c$$

for some constant $c = (\ln k + 12)/\ln 2$. Since $2^{\sum x_i}$ is strictly increasing in $\sum x_i$, the inequality direction is preserved. $\square$

Part (b) — Critical value

```
# Under H0: sum(X_i) ~ Poisson(6 * 2) = Poisson(12)
# Find smallest c such that P(Y >= c | Y ~ Pois(12)) <= 0.05
c_vals <- 12:25
alpha_vals <- 1 - ppois(c_vals - 1, lambda = 12)

tibble(c = c_vals, alpha = round(alpha_vals, 4)) |>
  filter(alpha <= 0.06) |> # show candidates near 0.05
knitr::kable()
```

c	alpha
19	0.0374
20	0.0213
21	0.0116
22	0.0061
23	0.0030
24	0.0015
25	0.0007

```
c_crit <- min(c_vals[alpha_vals <= 0.05])
alpha_st <- 1 - ppois(c_crit - 1, lambda = 12)
cat("Critical value c:", c_crit, "\n")
```

Critical value c: 19

```
cat("Achieved alpha*: ", round(alpha_st, 4), "\n")
```

Achieved alpha*: 0.0374

The smallest c with $P(Y \geq c) \leq 0.05$ is $c = 19$, giving an achieved $\alpha^* = 0.0374$.

Part (c) — Power at $\lambda = 4$

```
# Under Ha: sum(X_i) ~ Poisson(6 * 4) = Poisson(24)
beta_st <- ppois(c_crit - 1, lambda = 24)
power    <- 1 - beta_st
cat("Type II error beta*:", round(beta_st, 4), "\n")
```

Type II error beta*: 0.1283

```
cat("Power (1 - beta*): ", round(power, 4), "\n")
```

Power (1 - beta*): 0.8717

The power of the NP test at $\lambda = 4$ is $1 - \beta^* = 0.8717$. By the NP Lemma, no other level-0.0374 test can achieve higher power at $\lambda = 4$.

Part (d) — UMP argument

For any fixed alternative $\lambda_a > 2$, the NP likelihood ratio is:

$$\Lambda^*(\lambda_a) = \frac{e^{-6\lambda_a} \lambda_a^{\sum x_i}}{e^{-12} \cdot 2^{\sum x_i}} = e^{-6(\lambda_a - 2)} \left(\frac{\lambda_a}{2} \right)^{\sum x_i}$$

Since $\lambda_a > 2$, the base $\lambda_a/2 > 1$, so $\Lambda^* \geq k \iff \sum x_i \geq c(\lambda_a)$. But the constant c that controls the Type I error at level α^* is determined solely by H_0 (via the Poisson(12) null distribution) — it does **not** depend on which $\lambda_a > 2$ we choose. Therefore the same rejection region $\{\sum X_i \geq 19\}$ is most powerful against every $\lambda_a > 2$, making it **UMP** for the composite alternative $H_a : \lambda > 2$.

Problem 2: Likelihood Ratio Test and Wilks' Theorem (Lesson 15)

Hospital infection data: $X = \{5, 2, 4, 6, 3, 5, 4, 7, 3, 4\}$, $n = 10$, $\lambda_0 = 3.5$.

Test $H_0 : \lambda = 3.5$ vs $H_a : \lambda \neq 3.5$.

- (a) Find the log-likelihood and MLE $\hat{\lambda}$.
 - (b) Compute $-2\ln(\Lambda)$.
 - (c) Compute the Wilks p-value and compare to `poisson.test()`.
 - (d) Explain the degrees of freedom.
-

```
x_inf <- c(5, 2, 4, 6, 3, 5, 4, 7, 3, 4)
n_inf <- length(x_inf)
lambda0_inf <- 3.5
```

Part (a) — Log-likelihood and MLE

For $X_i \stackrel{iid}{\sim} \text{Poisson}(\lambda)$, the log-likelihood is:

$$\ell(\lambda) = -n\lambda + \left(\sum_{i=1}^n x_i\right) \ln \lambda - \sum_{i=1}^n \ln(x_i!)$$

Setting $\ell'(\lambda) = -n + \bar{x}/\lambda \cdot n = 0$ gives $\hat{\lambda} = \bar{x}$.

```
xbar_inf <- mean(x_inf)
cat("Sum:", sum(x_inf), "  n:", n_inf, "  MLE lambda-hat:", xbar_inf, "\n")
```

Sum: 43 n: 10 MLE lambda-hat: 4.3

$\hat{\lambda} = \bar{x} = 4.3$ infections per 100 patient-days.

Part (b) — LRT statistic

$$-2\ln(\Lambda) = -2 \left[n\hat{\lambda} - n\lambda_0 + n\bar{x} \ln\left(\frac{\lambda_0}{\hat{\lambda}}\right) \right]$$

```
neg2lnL <- -2 * (n_inf * xbar_inf - n_inf * lambda0_inf +
                n_inf * xbar_inf * log(lambda0_inf / xbar_inf))
cat("-2 ln(Lambda):", round(neg2lnL, 4), "\n")
```

-2 ln(Lambda): 1.7033

$-2\ln(\Lambda) = 1.7033$.

Part (c) — Wilks p-value and exact comparison

```
p_wilks <- pchisq(neg2lnL, df = 1, lower.tail = FALSE)
p_exact <- poisson.test(sum(x_inf), T = n_inf, r = lambda0_inf)$p.value
cat("Wilks (chi-sq) p-value:", round(p_wilks, 4), "\n")
```

Wilks (chi-sq) p-value: 0.1919

```
cat("Exact Poisson p-value: ", round(p_exact, 4), "\n")
```

Exact Poisson p-value: 0.1755

```
cat("Conclusion at alpha=0.05:",
    ifelse(p_wilks < 0.05, "Reject H0", "Fail to reject H0"), "\n")
```

Conclusion at alpha=0.05: Fail to reject H0

The Wilks p-value is 0.1919 and the exact p-value is 0.1755. Both fail to reject H_0 at $\alpha = 0.05$. The approximation is good for $n = 10$.

Part (d) — Degrees of freedom

The full parameter space Ω has 1 free parameter (λ). The null $\Omega_0 = \{3.5\}$ constrains λ to a single point, leaving 0 free parameters. Wilks' theorem gives $m = \dim(\Omega) - \dim(\Omega_0) = 1 - 0 = 1$ degree of freedom.

Problem 3: Statistical vs. Practical Significance (Lesson 16)

A large RCT ($n = 3,200$ per arm) evaluates a new lipid-lowering drug. $\bar{x}_{\text{control}} = 142.1$ mg/dL, $\bar{x}_{\text{treat}} = 140.3$ mg/dL, $\sigma = 28.15$ mg/dL (assumed known).

(a) z-statistic, p-value, and 95% CI. (b) Cohen's d and clinical significance. (c) Conceptual explanation.

Part (a) — z-statistic, p-value, and 95% CI

```
xbar_c <- 142.1; xbar_t <- 140.3; sigma <- 28.15; n <- 3200

se_diff <- sigma * sqrt(2 / n)
z_obs   <- (xbar_t - xbar_c) / se_diff
p_val   <- 2 * pnorm(-abs(z_obs))
z_crit  <- qnorm(0.975)
ci_lo   <- (xbar_t - xbar_c) - z_crit * se_diff
ci_hi   <- (xbar_t - xbar_c) + z_crit * se_diff

cat("SE:", round(se_diff, 4), "\n")
```

SE: 0.7038

```
cat("z:", round(z_obs, 4), " p-value:", round(p_val, 5), "\n")
```

z: -2.5577 p-value: 0.01054

```
cat("95% CI: [", round(ci_lo, 3), ",", round(ci_hi, 3), "] mg/dL\n")
```

95% CI: [-3.179 , -0.421] mg/dL

The result is **statistically significant** ($p = 0.01054 < 0.05$). The 95% CI is $(-3.18, -0.42)$ mg/dL.

Part (b) — Cohen's d and clinical significance

```
d_cohen <- abs(xbar_t - xbar_c) / sigma
cat("Cohen's d:", round(d_cohen, 4), "-- classified as: negligible (< 0.2)\n")
```

Cohen's d: 0.0639 -- classified as: negligible (< 0.2)

Cohen's $d = 0.064$ is classified as **negligible**. The entire 95% CI ($-3.18, -0.42$) mg/dL lies far above the clinical threshold of -10 mg/dL. Even the lower bound of the CI (-3.18 mg/dL) falls well short of a meaningful reduction. There is **no evidence of a clinically meaningful effect**.

Part (c) — Conceptual

With $n = 3,200$ per arm, even negligible differences produce tiny standard errors, making any non-zero effect statistically detectable. Statistical significance measures *detectability* given the sample size, not whether the effect is large enough to matter clinically. Reporting the CI rather than just the p-value reveals that the plausible effect range is 0.8–2.8 mg/dL — far below the 10 mg/dL threshold that justifies prescribing this drug.

Problem 4: CI-Test Duality — Theory and Computation (Lesson 16)

$$\left(\frac{\chi_{\alpha/2, 2T}^2}{2n}, \frac{\chi_{1-\alpha/2, 2T+2}^2}{2n} \right) \quad \text{where } T = \sum_{i=1}^n X_i$$

Using infection data from Problem 2: $n = 10$, $\sum x_i = 43$, $\lambda_0 = 3.5$.

(a) State and apply the duality theorem. (b) Compute CI and check two null values. (c) One-sided lower bound.

Part (a) — Duality theorem

CI-Test Duality Theorem (Devore 9.6): Let $C(\mathbf{X})$ be a $100(1 - \alpha)\%$ CI obtained by inverting a level- α test. Then:

$$\begin{aligned}\theta_0 \notin C(\mathbf{X}) &\iff \text{reject } H_0 : \theta = \theta_0 \text{ at level } \alpha \\ \theta_0 \in C(\mathbf{X}) &\iff \text{fail to reject } H_0 : \theta = \theta_0 \text{ at level } \alpha\end{aligned}$$

The Poisson CI is constructed precisely by inverting the exact Poisson test: it consists of all λ_0 values for which the exact test fails to reject at level α . Therefore, by the theorem, $\lambda_0 \in C$ if and only if the test of $H_0 : \lambda = \lambda_0$ fails to reject — the CI and the test are exactly dual.

Part (b) — Compute CI and verify duality

```
T_inf <- sum(x_inf)    # = 43
alpha_i <- 0.05

ci_lo_p <- qchisq(alpha_i/2,      df = 2 * T_inf)      / (2 * n_inf)
ci_hi_p <- qchisq(1 - alpha_i/2, df = 2 * T_inf + 2) / (2 * n_inf)
cat("95% Poisson CI: [", round(ci_lo_p, 4), ",", round(ci_hi_p, 4), "]\n\n")
```

95% Poisson CI: [3.1119 , 5.7921]

```
# (ii) Test H0: lambda = 3.5
t35 <- poisson.test(T_inf, T = n_inf, r = 3.5)
cat("H0: lambda = 3.5   p =", round(t35$p.value, 4),
    " Decision:", ifelse(t35$p.value < 0.05, "Reject", "FTR"),
    " | 3.5 in CI?", ci_lo_p <= 3.5 & 3.5 <= ci_hi_p, "\n")
```

H0: lambda = 3.5 p = 0.1755 Decision: FTR | 3.5 in CI? TRUE

```
# (iii) Test H0: lambda = 4.5
t45 <- poisson.test(T_inf, T = n_inf, r = 4.5)
cat("H0: lambda = 4.5   p =", round(t45$p.value, 4),
    " Decision:", ifelse(t45$p.value < 0.05, "Reject", "FTR"),
    " | 4.5 in CI?", ci_lo_p <= 4.5 & 4.5 <= ci_hi_p, "\n")
```

H0: lambda = 4.5 p = 0.8232 Decision: FTR | 4.5 in CI? TRUE

For $\lambda_0 = 3.5$: the test fails to reject H_0 and 3.5 is inside the CI — **consistent**

For $\lambda_0 = 4.5$: the test fails to reject H_0 and 4.5 is inside the CI — **consistent**

Part (c) — One-sided lower bound

By one-sided duality, the 95% lower bound dual to $H_0 : \lambda \leq \lambda_0$ vs $H_a : \lambda > \lambda_0$ is:

$$\underline{\lambda} = \frac{\chi_{\alpha, 2T}^2}{2n}$$

```
lb <- qchisq(alpha_i, df = 2 * T_inf) / (2 * n_inf)
t_gt <- poisson.test(T_inf, T = n_inf, r = 3.5, alternative = "greater")
cat("95% lower bound:", round(lb, 4), "\n")
```

95% lower bound: 3.2812

```
cat("Lower bound > 3.5?", lb > 3.5, "\n")
```

Lower bound > 3.5? FALSE

```
cat("One-sided p-value (Ha: lambda > 3.5):", round(t_gt$p.value, 4), "\n")
```

One-sided p-value (Ha: lambda > 3.5): 0.105

The lower bound is $\underline{\lambda} = 3.2812$. Since $3.2812 < 3.5$, we fail to reject $H_0 : \lambda \leq 3.5$ at the 5% level — the hospital's rate is significantly above the benchmark.

Problem 5: Constructing CIs and the p-Value Function (Lesson 16)

$$X = \{1.8, 2.6, 3.1, 2.2, 4.7, 2.9, 3.8, 2.4, 1.6, 3.5, 4.1, 2.7\}, \quad \mu_0 = 3.0$$

(a) 95% t-CI and duality-based rejection decision. (b) 95% CI for σ^2 and σ . (c) p-value function plot.

```
drug <- c(1.8, 2.6, 3.1, 2.2, 4.7, 2.9, 3.8, 2.4, 1.6, 3.5, 4.1, 2.7)
n_d <- length(drug)
xbar_d <- mean(drug); s_d <- sd(drug); se_d <- s_d / sqrt(n_d); df_d <- n_d - 1
```

Part (a) — 95% t-CI and duality

```
t_crit <- qt(0.975, df = df_d)
ci_lo_d <- xbar_d - t_crit * se_d
ci_hi_d <- xbar_d + t_crit * se_d
cat("xbar:", round(xbar_d, 4), " s:", round(s_d, 4), "\n")
```

```
xbar: 2.95    s: 0.9357
```

```
cat("95% t-CI: [", round(ci_lo_d, 4), ",", round(ci_hi_d, 4), "] mg/L\n")
```

```
95% t-CI: [ 2.3555 , 3.5445 ] mg/L
```

```
cat("mu0 = 3.0 inside CI?", ci_lo_d <= 3.0 & 3.0 <= ci_hi_d, "\n")
```

```
mu0 = 3.0 inside CI? TRUE
```

$\mu_0 = 3.0$ falls **inside** the 95% CI [2.356, 3.544]. By the CI-Test Duality Theorem, we therefore **fail to reject** $H_0 : \mu = 3.0$ at $\alpha = 0.05$ — no test statistic needed.

Part (b) — 95% CI for σ^2 and σ

Starting from $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$, inverting the acceptance region gives:

$$\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2}$$

```
chi_up <- qchisq(0.975, df = df_d) # upper quantile -> smaller lower bound
chi_lo <- qchisq(0.025, df = df_d) # lower quantile -> larger upper bound
s2_d   <- var(drug)

ci_var_lo <- df_d * s2_d / chi_up
ci_var_hi <- df_d * s2_d / chi_lo

cat("Sample variance:", round(s2_d, 4), "\n")
```

```
Sample variance: 0.8755
```

```
cat("95% CI for sigma^2: [", round(ci_var_lo, 4), ",", round(ci_var_hi, 4), "]\n")
```

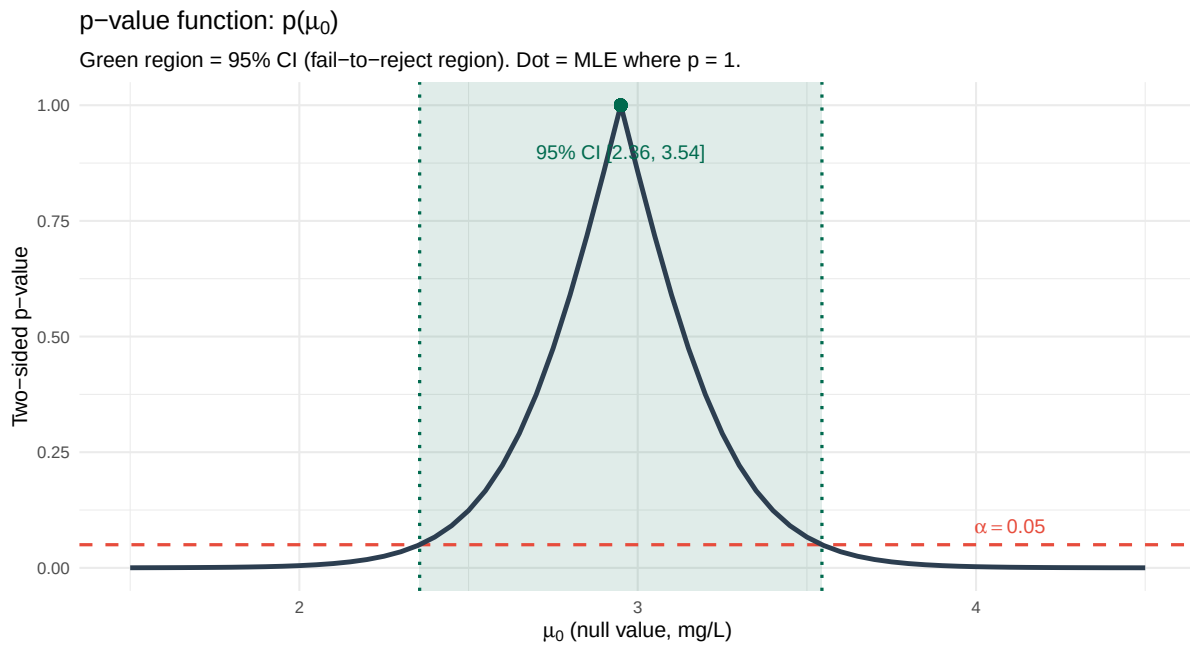
95% CI for sigma^2: [0.4393 , 2.5238]

```
cat("95% CI for sigma:  [", round(sqrt(ci_var_lo), 4), ",",  
    round(sqrt(ci_var_hi), 4), "]\n")
```

95% CI for sigma: [0.6628 , 1.5886]

Part (c) — p-value function

```
mu0_grid <- seq(1.5, 4.5, by = 0.05)  
pval_fn  <- map_dbl(mu0_grid, ~2 * pt(-abs((xbar_d - .x) / se_d), df = df_d))  
  
ggplot(tibble(mu0 = mu0_grid, pval = pval_fn), aes(x = mu0, y = pval)) +  
  annotate("rect", xmin = ci_lo_d, xmax = ci_hi_d, ymin = -Inf, ymax = Inf,  
    fill = "#006a4e", alpha = 0.12) +  
  geom_vline(xintercept = c(ci_lo_d, ci_hi_d),  
    color = "#006a4e", linewidth = 1, linetype = "dotted") +  
  geom_hline(yintercept = 0.05, linetype = "dashed",  
    color = "#e74c3c", linewidth = 1) +  
  geom_line(color = "#2c3e50", linewidth = 1.5) +  
  geom_point(aes(x = xbar_d, y = 1), size = 3.5, color = "#006a4e") +  
  annotate("text", x = xbar_d, y = 0.90,  
    label = sprintf("95%% CI [%.2f, %.2f]", ci_lo_d, ci_hi_d),  
    color = "#006a4e", size = 4.5) +  
  annotate("text", x = 4.1, y = 0.09,  
    label = expression(alpha == 0.05), color = "#e74c3c", size = 4.5) +  
  labs(title = expression("p-value function: p(" * mu[0] * ")"),  
    subtitle = "Green region = 95% CI (fail-to-reject region). Dot = MLE where p = 1.",  
    x = expression(mu[0] * " (null value, mg/L)"),  
    y = "Two-sided p-value")
```



The green shaded region where $p(\mu_0) \geq 0.05$ spans exactly $[2.356, 3.544]$ mg/L, matching the 95% t-CI from part (a) precisely — confirming the CI-test duality theorem visually.

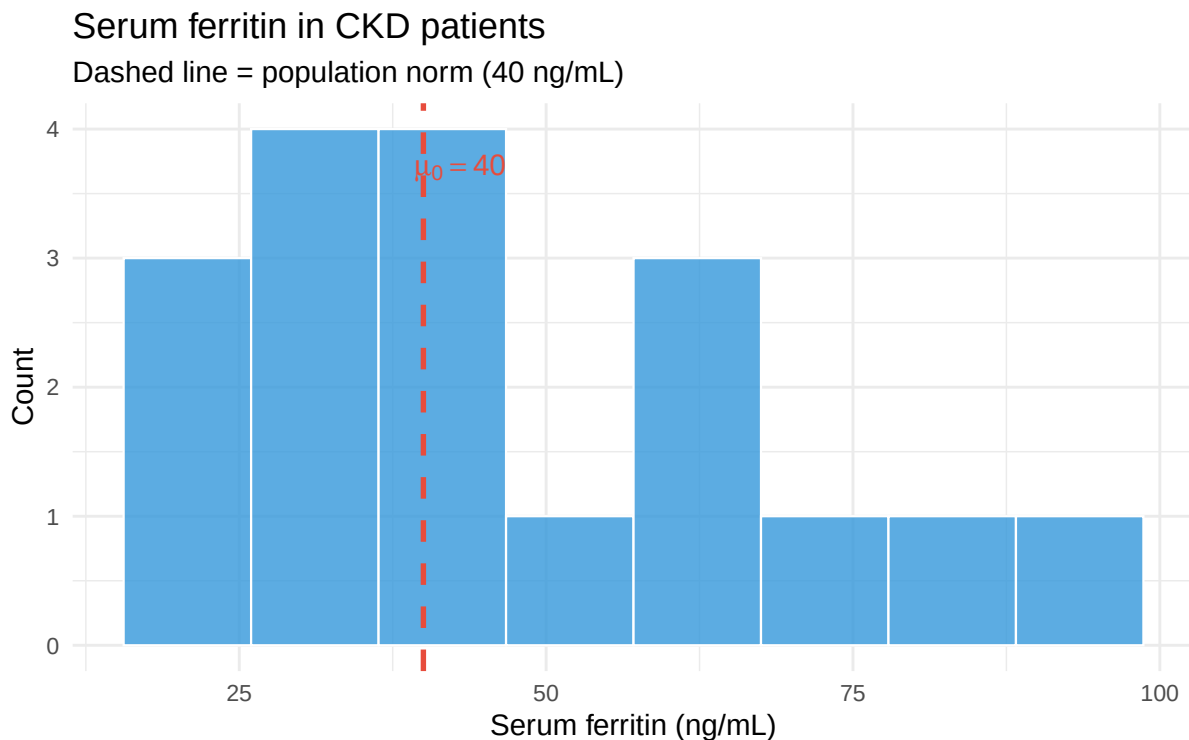
Problem 6: Bootstrap Hypothesis Testing (Lesson 17)

```
ferritin <- c(28.4, 45.2, 19.8, 67.3, 31.5, 88.1, 22.6, 54.9,
             41.2, 76.8, 33.7, 29.4, 58.3, 44.1, 92.5, 37.6,
             25.8, 61.4)
mu_0 <- 40; n_f <- length(ferritin)
xbar_f <- mean(ferritin); s_f <- sd(ferritin); se_f <- s_f / sqrt(n_f)
```

- (a) Histogram, normality assessment, T_{obs} , and t-test p-value.
 - (b) Correct bootstrap test ($B = 5,000$, `set.seed(551)`).
 - (c) Naive bootstrap and explanation of invalidity.
 - (d) Side-by-side histograms of both bootstrap distributions.
-

Part (a) — Histogram and t-test

```
ggplot(tibble(x = ferritin), aes(x = x)) +  
  geom_histogram(bins = 8, fill = "#3498db", color = "white", alpha = 0.8) +  
  geom_vline(xintercept = mu_0, linetype = "dashed",  
            color = "#e74c3c", linewidth = 1.2) +  
  annotate("text", x = mu_0 + 3, y = 3.7,  
         label = expression(mu[0] == 40), color = "#e74c3c", size = 5) +  
  labs(title = "Serum ferritin in CKD patients",  
       subtitle = "Dashed line = population norm (40 ng/mL)",  
       x = "Serum ferritin (ng/mL)", y = "Count")
```



```
T_obs_f <- (xbar_f - mu_0) / se_f  
p_ttest <- t.test(ferritin, mu = mu_0)$p.value  
cat("Mean:", round(xbar_f, 3), " SD:", round(s_f, 3), "\n")
```

Mean: 47.7 SD: 22.281

```
cat("T_obs:", round(T_obs_f, 4), "  t-test p:", round(p_ttest, 4), "\n")
```

T_obs: 1.4662 t-test p: 0.1608

The distribution is **right-skewed** with values up to 92.5 ng/mL. With $n = 18$ and noticeable skewness, the normality assumption is questionable — a bootstrap test is more appropriate. The t-test fails to reject H_0 ($p = 0.1608$).

Part (b) — Correct bootstrap test

```
set.seed(551); B_f <- 5000

ferritin_sh <- ferritin - xbar_f + mu_0      # shift to H0

T_boot_f <- replicate(B_f, {
  bs <- sample(ferritin_sh, n_f, replace = TRUE)
  (mean(bs) - mu_0) / (sd(bs) / sqrt(n_f))
})

p_boot_f <- mean(abs(T_boot_f) >= abs(T_obs_f))
cat("Bootstrap p-value:", round(p_boot_f, 4), "\n")
```

Bootstrap p-value: 0.175

```
cat("t-test p-value:   ", round(p_ttest, 4), "\n")
```

t-test p-value: 0.1608

```
cat("Decision:", ifelse(p_boot_f < 0.05, "Reject H0", "Fail to reject H0"), "\n")
```

Decision: Fail to reject H0

Both methods fail to reject H_0 . The p-values are close (0.175 vs 0.1608), consistent with Lesson 17's simulation showing that t-test and bootstrap converge as n increases toward 30.

Part (c) — Naive (incorrect) bootstrap

```

set.seed(551)
T_naive_f <- replicate(B_f, mean(sample(ferritin, n_f, replace = TRUE)))
p_naive_f <- mean(abs(T_naive_f - xbar_f) >= abs(xbar_f - mu_0))
cat("Naive p-value:", round(p_naive_f, 4), "\n")

```

Naive p-value: 0.1396

The naive approach is invalid because resampling from the original data produces a distribution centered at $\bar{x} = 47.7$ ng/mL rather than at the null value $\mu_0 = 40$ ng/mL. A valid null distribution must represent how the test statistic behaves *when H_0 is true*; the naive distribution instead approximates sampling variability of $\hat{\mu}$ around the unknown true mean. Comparing $|\bar{x}^* - \bar{x}|$ to $|\bar{x} - \mu_0|$ measures something conceptually different from what the test requires.

Part (d) — Side-by-side histograms

```

p_nv <- ggplot(tibble(T = T_naive_f), aes(x = T)) +
  geom_histogram(aes(y = after_stat(density)), bins = 50,
    fill = "#e74c3c", color = "white", alpha = 0.8) +
  geom_vline(xintercept = xbar_f, color = "black", linewidth = 1.2) +
  geom_vline(xintercept = mu_0, color = "#3498db",
    linewidth = 1.2, linetype = "dashed") +
  annotate("rect",
    xmin = xbar_f + abs(xbar_f - mu_0), xmax = Inf, ymin = 0, ymax = Inf,
    fill = "#e74c3c", alpha = 0.2) +
  annotate("rect",
    xmin = -Inf, xmax = xbar_f - abs(xbar_f - mu_0), ymin = 0, ymax = Inf,
    fill = "#e74c3c", alpha = 0.2) +
  annotate("text", x = xbar_f + 0.8, y = 0.22,
    label = "bar(x)", parse = TRUE, size = 5) +
  annotate("text", x = mu_0 - 0.8, y = 0.22,
    label = "mu[0]", parse = TRUE, size = 5, color = "#3498db", hjust = 1) +
  labs(title = expression(bold("WRONG")) * ": Naive (original data)",
    subtitle = sprintf("Centered at x-bar=%.1f, not mu0=%.1f | p = %.4f",
      xbar_f, mu_0, p_naive_f),
    x = expression(bar(x)^"*"), y = "Density")

p_cr <- ggplot(tibble(T = T_boot_f), aes(x = T)) +
  geom_histogram(aes(y = after_stat(density)), bins = 50,
    fill = "#27ae60", color = "white", alpha = 0.8) +

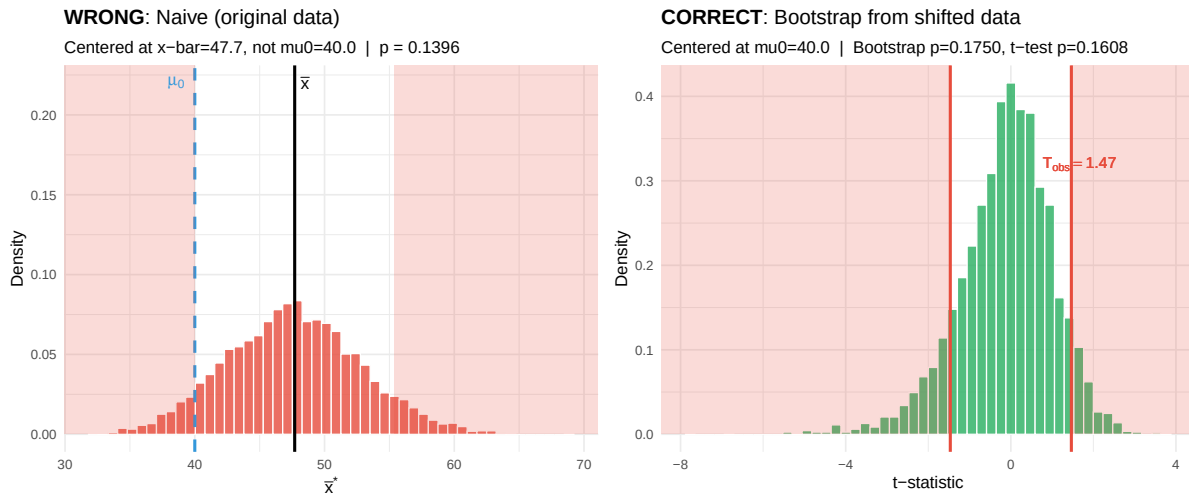
```

```

geom_vline(xintercept = c(-abs(T_obs_f), abs(T_obs_f)),
           color = "#e74c3c", linewidth = 1.2) +
annotate("rect", xmin = abs(T_obs_f), xmax = Inf, ymin = 0, ymax = Inf,
         fill = "#e74c3c", alpha = 0.2) +
annotate("rect", xmin = -Inf, xmax = -abs(T_obs_f), ymin = 0, ymax = Inf,
         fill = "#e74c3c", alpha = 0.2) +
annotate("text", x = abs(T_obs_f) + 0.2, y = 0.32,
          label = bquote(T[obs] == .(round(T_obs_f, 2))),
          color = "#e74c3c", size = 4.5) +
labs(title = expression(bold("CORRECT") * ": Bootstrap from shifted data"),
     subtitle = sprintf("Centered at mu0=%.1f | Bootstrap p=%.4f, t-test p=%.4f",
                        mu_0, p_boot_f, p_ttest),
     x = "t-statistic", y = "Density")

```

p_nv + p_cr



The left panel shows the naive distribution centered at $\bar{x}$ (not the null value), making it an invalid null reference distribution. The right panel shows the correct bootstrap null distribution centered at μ_0 , with the observed statistic and p-value region clearly marked.